Stochastic Perturbed Proximal Augmented Lagrangian for Nonconvex Constrained Optimization

April 12, 2026

1 Introduction

We consider the following nonconvex stochastic optimization problem with mixed deterministic constraints:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathcal{X}} \quad & F(\boldsymbol{x}) := f(\boldsymbol{x}) + g(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}) = \mathbf{0}, \\ & \boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}) \le \mathbf{0}, \end{aligned} \tag{\textbf{P}}$$

where $\mathcal{E}$ and $\mathcal{I}$ index the equality and inequality blocks, respectively, and $\boldsymbol{c}(\boldsymbol{x}) := \big(\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}), \boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x})\big) \in \mathbb{R}^m$ with $m := |\mathcal{E}| + |\mathcal{I}|$. Here $f(\boldsymbol{x}) = \mathbb{E}_{\xi}[\mathbf{f}(\boldsymbol{x}, \xi)]$ is a smooth nonconvex expectation objective, $g : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a proper closed proximal-friendly function, and the mappings $\boldsymbol{c}_{\mathcal{E}} : \mathbb{R}^n \to \mathbb{R}^{|\mathcal{E}|}$ and $\boldsymbol{c}_{\mathcal{I}} : \mathbb{R}^n \to \mathbb{R}^{|\mathcal{I}|}$ are continuously differentiable. The set $\mathcal{X} \subseteq \mathbb{R}^n$ is nonempty and closed, the feasible set of (**P**) is nonempty, and F is bounded below by $F^* > -\infty$. Exact evaluation of ∇f is often impractical, so we assume access only to stochastic gradients queried at sample points.

A key obstacle in this setting is that equality and inequality constraints enter augmented-Lagrangian models asymmetrically: equality blocks generate the standard quadratic augmentation, whereas inequality blocks require one-sided activation. Our approach is to absorb this asymmetry into a single *effective multiplier*, which leads to a compact perturbed proximal augmented Lagrangian with a fixed penalty parameter and a genuinely single-loop primal–dual update.

1.1 Related Work

Deterministic nonconvex constrained optimization. For deterministic nonconvex problems with equality and inequality constraints, first-order methods with non-asymptotic complexity guarantees have been studied extensively. Proximal point methods [4, 20] achieve $\mathcal{O}(\varepsilon^{-3})$ iteration complexity for smooth objectives via strongly convex subproblems, and Boob et al. [5] further attain $\mathcal{O}(\varepsilon^{-2})$ by constructing quadratic models with adaptive constraint levels. Among penalty and augmented Lagrangian methods (ALMs), most approaches require the penalty parameter to grow as $\Omega(\varepsilon^{-1})$, yielding $\mathcal{O}(\varepsilon^{-3})$ iteration complexity [15, 13]. Xie and Wright [24] achieve $\mathcal{O}(\varepsilon^{-2})$ outer-loop complexity with a proximal ALM under a constant merit parameter, but the algorithm is double-loop and the merit parameter must satisfy $\rho = \Omega(\varepsilon^{-\eta})$ for some $\eta \in (0, 1)$, making it ε-dependent in theory. SQP methods [9, 6] also attain $\mathcal{O}(\varepsilon^{-2})$ complexity.

Stochastic nonconvex constrained optimization. We organize the literature by the two canonical settings: *semi-stochastic* (stochastic objective, deterministic constraints) and *fully stochastic* (both objective and constraints involve stochastic oracles).

Semi-stochastic problems. Without additional structure on the noise, proximal point [4, 20], penalty [14], and SQP [9] methods all achieve $\mathcal{O}(\varepsilon^{-4})$ oracle complexity in stochastic gradient evaluations, matching the $\mathcal{O}(\varepsilon^{-4})$ lower bound for unconstrained stochastic optimization [3]. Under a mean-squared smoothness

(MSS) condition, variance reduction techniques—STORM [10] and SPIDER [11]—improve the complexity to $\mathcal{O}(\varepsilon^{-3})$. Representative results include the variance-reduced linearized ALM of Shi et al. [23], the truncated variance reduction approach of Lu et al. [18], and the inexact ALM of Li et al. [15], each attaining $\mathcal{O}(\varepsilon^{-3})$ oracle complexity under MSS for nonlinear constraints.

Fully stochastic problems. When constraint evaluations are also stochastic, complexity results are uniformly higher. Proximal point methods [4, 20] incur $\mathcal{O}(\varepsilon^{-6})$ total oracle complexity due to the cost of solving stochastic strongly convex subproblems. Penalty-based ALMs employing STORM-type variance reduction [1, 7] face a large merit parameter $\beta = \Theta(\varepsilon^{-1})$, which amplifies the variance of the stochastic gradient and yields $\mathcal{O}(\varepsilon^{-5})$ complexity even under MSS. Exact penalty methods combined with SPIDER [17] achieve $\mathcal{O}(\varepsilon^{-4})$ (sub)gradient and $\mathcal{O}(\varepsilon^{-6})$ constraint-evaluation complexity for nonsmooth problems. Cui et al. [8] recently propose an adaptive exact penalty method for smooth *equality*-constrained problems, obtaining $\mathcal{O}(\varepsilon^{-3})$ gradient and $\mathcal{O}(\varepsilon^{-5})$ constraint-evaluation complexity; however, the analysis is confined to equality constraints and relies on an exact penalty reformulation. Stochastic SQP for equality-constrained fully stochastic problems [21] currently achieves only $\mathcal{O}(\varepsilon^{-8})$.

ALMs with fixed penalty parameter. A fundamental challenge in ALM design for nonlinear constraints is that standard convergence analysis requires the penalty parameter to scale as $\Omega(\varepsilon^{-1})$. This growth inflates the Lipschitz constant of the augmented Lagrangian and forces the step size to shrink proportionally, degrading the stochastic oracle complexity. For polyhedral (linear) constraints, the Luo–Pang global error bound [19] decouples feasibility control from the penalty magnitude. Exploiting this structure, Zhang and Luo [25] achieve $\mathcal{O}(\varepsilon^{-2})$ iteration complexity for deterministic nonconvex problems under a genuinely constant penalty. In the stochastic setting, Alacaoglu and Wright [2] combine a fixed penalty with variance reduction to attain the optimal $\mathcal{O}(\varepsilon^{-3})$ sample complexity for linear equality constraints. Huang et al. [12] extend the fixed-penalty result to linear inequality constraints via Moreau envelope smoothing, also reaching $\mathcal{O}(\varepsilon^{-3})$. For general nonlinear constraints, existing stochastic methods either require an increasing penalty parameter [16, 2], yielding $\mathcal{O}(\varepsilon^{-4})$ oracle complexity, or introduce dual momentum to partially compensate, achieving $\mathcal{O}(\varepsilon^{-3.5})$ [22]. To date, no single-loop ALM with a genuinely constant penalty parameter has been shown to achieve $\mathcal{O}(\varepsilon^{-3})$ oracle complexity for general nonlinear mixed constraints in the stochastic setting.

Our perturbed ALM addresses this gap by combining a uniform proximal perturbation on the multiplier vector with an effective-multiplier map that absorbs the equality/inequality asymmetry. This yields a compact PPAL model, keeps the penalty parameter fixed, and supports a single-loop stochastic scheme for nonlinear mixed constraints.

1.2 Our Contributions

- **Unified PPAL formulation for mixed constraints.** We introduce an effective multiplier $\hat{\boldsymbol{\lambda}}(\boldsymbol{\lambda}, \boldsymbol{x})$ that absorbs the difference between equality and inequality blocks, allowing the perturbed proximal augmented Lagrangian to be written in a single compact form with one fixed penalty parameter $\rho > 0$.
- **Single-loop stochastic primal–dual algorithm.** We develop a Stochastic Perturbed Proximal Augmented Lagrangian (SPPAL) method that combines STORM variance reduction, a damped auxiliary multiplier update, and a closed-form blockwise multiplier step that handles equality and inequality constraints in a unified way.
- **Residual-based convergence interface.** We formulate mixed-constraint KKT residuals consistent with the effective multiplier and establish the analysis interface needed for fixed-penalty complexity bounds and feasibility control in the nonlinear stochastic setting.

1.3 Notation

Throughout this paper, $\|\cdot\|$ denotes the Euclidean norm and $\langle\cdot,\cdot\rangle$ denotes the Euclidean inner product. For $a \in \mathbb{R}$, let $[a]_+ := \max\{a, 0\}$, and apply $[\cdot]_+$ to vectors componentwise. We use $\mathbb{E}_\xi[\cdot]$ for expectation

with respect to ξ, let $[p] := \{1, \ldots, p\}$ and $\mathbb{N} := \{0, 1, 2, \ldots\}$, and write $\overline{\mathbb{R}} := \mathbb{R} \cup \{+\infty\}$. The Jacobian of $\boldsymbol{c}$ is denoted by $J_{\boldsymbol{c}}(\boldsymbol{x}) := [J_{\boldsymbol{c}_\mathcal{E}}(\boldsymbol{x}), J_{\boldsymbol{c}_\mathcal{I}}(\boldsymbol{x})] \in \mathbb{R}^{n\times m}$. For $X \subseteq \mathbb{R}^n$, δ_X denotes its indicator function and $\operatorname{dist}(\boldsymbol{x}, X) := \inf_{\boldsymbol{y}\in X} \|\boldsymbol{x} - \boldsymbol{y}\|$. For complexity notation, we write $K = \mathcal{O}(\varepsilon)$ if $K \le B\varepsilon$ for some $B > 0$, and $K = \Theta(\varepsilon)$ if $b\varepsilon \le K \le B\varepsilon$ for some constants $0 < b < B < +\infty$.

1.4 Organization

The rest of the paper is organized as follows. Section 2 introduces the assumptions and the stochastic KKT criterion. Section 3 presents the perturbed proximal ALM formulation and the single-loop algorithm. Section 4 establishes stochastic complexity and residual bounds. Section 5 reports numerical experiments. Additional technical lemmas are provided in Appendix A.

2 Preliminaries and Assumptions

2.1 Main Assumptions and stationarity

In this subsection, we formally state our assumptions for Problem (**P**) and the stationarity for problem.

Assumption 1 *Function $f(\cdot)$ is L_f-smooth over $\mathcal{X}$. The mapping $\boldsymbol{c} = (\boldsymbol{c}_\mathcal{E}, \boldsymbol{c}_\mathcal{I}) : \mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable over $\mathcal{X}$, and there exist constants $C, G, L_c > 0$ such that, for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$,*

$$\|\boldsymbol{c}(\boldsymbol{x})\| \le C, \qquad \|J_{\boldsymbol{c}}(\boldsymbol{x})\| \le G, \qquad \|J_{\boldsymbol{c}}(\boldsymbol{x}) - J_{\boldsymbol{c}}(\boldsymbol{y})\| \le L_c\|\boldsymbol{x} - \boldsymbol{y}\|. \tag{2.1}$$

Assumption 2 *For any $\boldsymbol{x}$, there exists $\sigma > 0$ such that*

$$\begin{aligned} &\mathbb{E}_\xi[\nabla \mathbf{f}(\boldsymbol{x}, \boldsymbol{\xi})] = \nabla f(\boldsymbol{x}), \\ &\|\nabla \mathbf{f}(\boldsymbol{x}, \boldsymbol{\xi}) - \nabla f(\boldsymbol{x})\| \le \sigma, \quad \forall \xi \end{aligned} \tag{2.2}$$

We also introduce a standard assumption, also known as the mean-squared smoothness assumption, widely used in variance reduction methods.

Assumption 3 *For almost every $\boldsymbol{\xi} \in \Xi$, $f(\cdot, \boldsymbol{\xi})$ is differentiable and for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$,*

$$\mathbb{E}_{\boldsymbol{\xi}}\left[\|\nabla f(\boldsymbol{x}; \boldsymbol{\xi}) - \nabla f(\boldsymbol{y}; \boldsymbol{\xi})\|^2\right] \le L_f^2\|\boldsymbol{x} - \boldsymbol{y}\|^2.$$

Assumption 4 (Boundedness of Multipliers) *The sequence of dual variables $\{\boldsymbol{\lambda}^k\}$ generated by Algorithm 1 is contained in a convex compact set $\Lambda \subset \mathbb{R}^{|\mathcal{E}|} \times \mathbb{R}_+^{|\mathcal{I}|}$, i.e., there exists a constant $B_\lambda > 0$ such that*

$$\|\boldsymbol{\lambda}^k\| \le B_\lambda, \quad \forall\, k \ge 0. \tag{2.3}$$

2.2 Stochastic KKT Point

Definition 1 (Stochastic KKT point) *A point $\boldsymbol{x}^* \in \mathcal{X}$ is called a* stationary point *of Problem (**P**) if there exists $\boldsymbol{\lambda}^* \in \mathbb{R}^{|\mathcal{E}|} \times \mathbb{R}_+^{|\mathcal{I}|}$ such that*

$$\begin{aligned} \mathbf{0} &\in \nabla f(\boldsymbol{x}^*) + \partial g(\boldsymbol{x}^*) + J_{\boldsymbol{c}_\mathcal{E}}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}_\mathcal{E}^* + J_{\boldsymbol{c}_\mathcal{I}}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}_\mathcal{I}^* + \mathcal{N}_\mathcal{X}(\boldsymbol{x}^*), \\ \boldsymbol{c}_\mathcal{E}(\boldsymbol{x}^*) &= \mathbf{0}, \qquad \boldsymbol{c}_\mathcal{I}(\boldsymbol{x}^*) \le \mathbf{0}, \qquad \boldsymbol{\lambda}_\mathcal{I}^* \odot \boldsymbol{c}_\mathcal{I}(\boldsymbol{x}^*) = \mathbf{0}. \end{aligned} \tag{2.4}$$

Definition 2 (ε-KKT point) *A point $\boldsymbol{x} \in \mathcal{X}$ is called an* ε-KKT point *of Problem (**P**) if there exists $\boldsymbol{\lambda} \in \mathbb{R}^{|\mathcal{E}|} \times \mathbb{R}_+^{|\mathcal{I}|}$ such that*

$$\begin{aligned} \operatorname{dist}\Big(\mathbf{0},\ \nabla f(\boldsymbol{x}) + \partial g(\boldsymbol{x}) + \mathcal{N}_\mathcal{X}(\boldsymbol{x}) + J_{\boldsymbol{c}_\mathcal{E}}(\boldsymbol{x})^\top \boldsymbol{\lambda}_\mathcal{E} + J_{\boldsymbol{c}_\mathcal{I}}(\boldsymbol{x})^\top \boldsymbol{\lambda}_\mathcal{I}\Big) &\le \varepsilon, \\ \|\boldsymbol{c}_\mathcal{E}(\boldsymbol{x})\| \le \varepsilon, \qquad \|[\boldsymbol{c}_\mathcal{I}(\boldsymbol{x})]_+\| &\le \varepsilon, \\ \sum_{i\in\mathcal{I}} |\boldsymbol{\lambda}_i c_i(\boldsymbol{x})| &\le \varepsilon. \end{aligned} \tag{2.5}$$

We measure progress by the mixed-constraint residual

$$R(\boldsymbol{x},\boldsymbol{\lambda}) := \operatorname{dist}\Big(-\nabla f(\boldsymbol{x}) - J_{\boldsymbol{c}}(\boldsymbol{x})^\top \hat{\boldsymbol{\lambda}}(\boldsymbol{\lambda},\boldsymbol{x}),\ \partial g(\boldsymbol{x}) + \mathcal{N}_{\mathcal{X}}(\boldsymbol{x})\Big)^2 + \|\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x})\|^2 + \|[\boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x})]_+\|^2. \tag{2.6}$$

3 Methodology and Algorithm

To absorb the structural difference between equality and inequality constraints, we define the *effective multiplier*

$$\hat{\boldsymbol{\lambda}}_i(\boldsymbol{\lambda},\boldsymbol{x}) := \begin{cases} \boldsymbol{\lambda}_i + \rho\, c_i(\boldsymbol{x}), & i \in \mathcal{E}, \\ \big[\boldsymbol{\lambda}_i + \rho\, c_i(\boldsymbol{x})\big]_+, & i \in \mathcal{I}. \end{cases} \tag{3.1}$$

and write $\hat{\boldsymbol{\lambda}}(\boldsymbol{\lambda},\boldsymbol{x}) \in \mathbb{R}^m$ for the stacked vector. This map absorbs the equality/inequality asymmetry, while the perturbation term acts on the full multiplier vector symmetrically.

$$\mathcal{L}_\rho(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\mu}) = \underbrace{f(\boldsymbol{x}) + \frac{1}{2\rho}\|\hat{\boldsymbol{\lambda}}(\boldsymbol{\lambda},\boldsymbol{x})\|^2 - \frac{1}{2\rho}\|\boldsymbol{\lambda}\|^2}_{\varphi_\rho(\boldsymbol{x},\boldsymbol{\lambda})} + g(\boldsymbol{x}) - \frac{1}{2\rho}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2. \tag{3.2}$$

Remark 3.1 *For each $i \in \mathcal{E}$,*

$$\frac{1}{2\rho}\big(\boldsymbol{\lambda}_i + \rho c_i(\boldsymbol{x})\big)^2 - \frac{1}{2\rho}\boldsymbol{\lambda}_i^2 = \boldsymbol{\lambda}_i c_i(\boldsymbol{x}) + \frac{\rho}{2} c_i(\boldsymbol{x})^2.$$

For each $i \in \mathcal{I}$,

$$\frac{1}{2\rho}\big[\boldsymbol{\lambda}_i + \rho c_i(\boldsymbol{x})\big]_+^2 - \frac{1}{2\rho}\boldsymbol{\lambda}_i^2$$

recovers the sharp augmented-Lagrangian term for inequality constraints. Hence $\hat{\boldsymbol{\lambda}}(\boldsymbol{\lambda},\boldsymbol{x})$ absorbs the block asymmetry, while $-\frac{1}{2\rho}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$ supplies a uniform proximal perturbation.

$$\nabla_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{x},\boldsymbol{\lambda}) = \nabla f(\boldsymbol{x}) + J_{\boldsymbol{c}}(\boldsymbol{x})^\top \hat{\boldsymbol{\lambda}}(\boldsymbol{\lambda},\boldsymbol{x}). \tag{3.3}$$

3.1 Algorithm

Stochastic gradient estimator. We replace $\nabla f(\boldsymbol{x}^k)$ with the STORM variance-reduced estimator:

$$\boldsymbol{d}^k = \nabla \mathbf{f}(\boldsymbol{x}^k,\xi^k) + (1-\gamma_{k-1})\big(\boldsymbol{d}^{k-1} - \nabla \mathbf{f}(\boldsymbol{x}^{k-1},\xi^k)\big), \tag{3.4}$$

where ξ^k is an independent sample drawn at iteration k.

Update of $\boldsymbol{x}$.

$$\boldsymbol{x}^{k+1} \in \operatorname*{argmin}_{\boldsymbol{x}\in\mathcal{X}} \left\{ g(\boldsymbol{x}) + \left\langle \hat{\nabla}_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k),\ \boldsymbol{x}-\boldsymbol{x}^k \right\rangle + \frac{1}{2\eta}\|\boldsymbol{x}-\boldsymbol{x}^k\|^2 \right\}, \tag{3.5}$$

where $\hat{\nabla}_{\boldsymbol{x}}\varphi_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k) := \boldsymbol{d}^k + J_{\boldsymbol{c}}(\boldsymbol{x}^k)^\top \hat{\boldsymbol{\lambda}}(\boldsymbol{\lambda}^k,\boldsymbol{x}^k)$.

Update of $\boldsymbol{\mu}$.

$$\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \theta_k\big(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\big), \tag{3.6}$$

with a diminishing step-size

$$\theta_k = \min\left\{\theta_0,\ \frac{\rho\delta_k}{\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 + 1}\right\}, \tag{3.7}$$

and δ_k satisfies

$$\delta_0 \in (0,1], \qquad \lim_{k\to\infty}\delta_k = 0, \qquad \text{and} \qquad \sum_{k=0}^{\infty}\delta_k = +\infty. \tag{3.8}$$

Update of $\boldsymbol{\lambda}$.

$$\boldsymbol{\lambda}^{k+1} = \hat{\boldsymbol{\lambda}}(\boldsymbol{\mu}^{k+1}, \boldsymbol{x}^{k+1}). \tag{3.9}$$

Equivalently,

$$\begin{aligned}\boldsymbol{\lambda}_{\mathcal{E}}^{k+1} &= \boldsymbol{\mu}_{\mathcal{E}}^{k+1} + \rho\, \boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}^{k+1}),\\ \boldsymbol{\lambda}_{\mathcal{I}}^{k+1} &= \big[\boldsymbol{\mu}_{\mathcal{I}}^{k+1} + \rho\, \boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}^{k+1})\big]_+.\end{aligned}$$

If $\boldsymbol{\lambda}_{\mathcal{I}}^0, \boldsymbol{\mu}_{\mathcal{I}}^0 \geq \mathbf{0}$, then $\boldsymbol{\lambda}_{\mathcal{I}}^k, \boldsymbol{\mu}_{\mathcal{I}}^k \geq \mathbf{0}$ for all k.

Algorithm 1 Stochastic Perturbed Proximal Augmented Lagrangian (SPPAL)

1: **Input:** $(\boldsymbol{x}^0, \boldsymbol{\lambda}^0, \boldsymbol{\mu}^0)$ with $\boldsymbol{\lambda}_{\mathcal{I}}^0, \boldsymbol{\mu}_{\mathcal{I}}^0 \geq \mathbf{0}$, parameters $\eta > 0$, $\rho > 0$, $\theta_0 \in (0, 1]$, momentum sequence $\{\gamma_k\}$, and damping sequence $\{\delta_k\}$ satisfying $\delta_k \to 0$ and $\sum_k \delta_k = +\infty$.
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Update $\boldsymbol{x}^{k+1}$ by (3.5).
4: Update $\boldsymbol{\mu}^{k+1}$ by (3.6).
5: Update $\boldsymbol{\lambda}^{k+1}$ by (3.9).
6: **end for**
7: **Output:** $\boldsymbol{x}^R$ (randomly selected iterate).

4 Stochastic Complexity Analysis

Lemma 1 (Smoothness of φ_ρ) *Under Assumptions 1 and 4, the function $\varphi_\rho(\cdot, \boldsymbol{\lambda})$ is L_φ-smooth for any $\boldsymbol{\lambda}$ with $\|\boldsymbol{\lambda}\| \leq B_\lambda$, i.e.,*

$$\|\nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) - \nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{y}, \boldsymbol{\lambda})\| \leq L_\varphi \|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}, \tag{4.1}$$

where

$$L_\varphi := L_f + L_c(B_\lambda + \rho C) + \rho G^2. \tag{4.2}$$

Lemma 2 (Basic relations on $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$) *Let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumption 1, the following relations hold for all $k \geq 0$:*

$$\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|^2 = \theta_k^2 \|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 \leq \frac{\rho^2 \delta_k^2}{4}, \tag{4.3}$$

$$\theta_k \|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 \leq \rho \delta_k, \tag{4.4}$$

$$\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\lambda}^k\|^2 = (1 - \theta_k)^2 \|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2, \tag{4.5}$$

$$\mathbb{E}[\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2] \leq 2\rho^2 L_c^2\, \mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \frac{\rho^2 \delta_k^2}{2}. \tag{4.6}$$

Lemma 3 (Sufficient Descent) *Let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumptions 1, 2, and 4, the following inequality holds for all $k \geq 0$:*

$$\mathbb{E}\big[\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{\mu}^{k+1}) \,\big|\, \mathcal{F}_k\big] \;\leq\; \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) - C_1\, \mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \,\big|\, \mathcal{F}_k\big] + \frac{\eta}{2}\|\boldsymbol{e}^k\|^2 + \hat{\delta}_k, \tag{4.7}$$

where

$$C_1 := \frac{1}{2\eta} - \frac{L_\varphi}{2} - \rho L_c^2 > 0, \qquad \hat{\delta}_k := \frac{\rho \delta_k^2}{4} + \delta_k. \tag{4.8}$$

Furthermore, rearranging (4.7) gives

$$C_1\, \mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k\big] \leq \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) - \mathbb{E}\big[\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{\mu}^{k+1}) \mid \mathcal{F}_k\big] + \frac{\eta}{2}\|\boldsymbol{e}^k\|^2 + \hat{\delta}_k. \tag{4.9}$$

Lemma 4 (Primal Stationarity Residual Bound) *Let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumptions 1 and 2, define the stationarity residual*

$$\boldsymbol{\zeta}_{\boldsymbol{x}}^{k+1} \in \nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}) + \partial g(\boldsymbol{x}^{k+1}) + \mathcal{N}_{\mathcal{X}}(\boldsymbol{x}^{k+1}). \tag{4.10}$$

Then for all $k \geq 0$, Lemma 2 (4.6),

$$\mathbb{E}\big[\|\boldsymbol{\zeta}_{\boldsymbol{x}}^{k+1}\|^2\big] \;\leq\; 4\left[\left(L_\varphi + \frac{1}{\eta}\right)^2 + 2G^2\rho^2 L_c^2\right] \mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2\big] \;+\; 4\,\mathbb{E}\big[\|\boldsymbol{e}^k\|^2\big] \;+\; 2G^2\rho^2\delta_k^2. \tag{4.11}$$

where $\boldsymbol{e}^k := \boldsymbol{d}^k - \nabla f(\boldsymbol{x}^k)$ is the STORM estimation error.

Lemma 5 (Recursive Gradient Error) *Suppose that Assumptions 1, 2, and 3 hold. Then, for any $k \geq 0$, it holds that*

$$\mathbb{E}\big[\|\boldsymbol{e}^{k+1}\|^2\big] \leq (1-\gamma_k)^2\,\mathbb{E}\big[\|\boldsymbol{e}^k\|^2\big] + 2(1-\gamma_k)^2 L_f^2\,\mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2\big] + 2\gamma_k^2\sigma^2. \tag{4.12}$$

Lemma 6 *Under Assumptions 1–3, let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be generated by Algorithm 1 with non-increasing step sizes $\{\eta_k\}$ satisfying*

$$(1 - L_{c,\varphi}\eta_k)^{-1} \leq 2 \tag{4.13}$$

where $L_{c,\varphi} = L_\varphi + 2\rho L_c^2$. Then for any $K \geq 1$,

$$\frac{1}{K}\sum_{k=0}^{K-1} \gamma_k\,\mathbb{E}\big[\|\boldsymbol{e}^k\|^2\big] \leq \frac{16\eta_1 L_f^2\big(\mathcal{L}_\rho^0 - \mathcal{L}_\rho^*\big)}{K} + \frac{2\,\mathbb{E}\big[\|\boldsymbol{e}^1\|^2\big]}{K} + \frac{16\eta_1 L_f^2}{K}\sum_{k=0}^{K-1}\hat{\delta}_k + \frac{1}{K}\sum_{k=0}^{K-1} 4\gamma_k^2\sigma^2, \tag{4.14}$$

where $\mathcal{L}_\rho^* \geq F^*$ is a lower bound on $\mathcal{L}_\rho$.

Lemma 7 (KKT residual stationarity)

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[\|\zeta_x^{k+1}\|^2\big]$$

$$\leq \left(\frac{\eta_1 L_f^2}{\gamma_K \cdot K} + \frac{1}{\eta_k \cdot K}\right)\big(\mathcal{L}_\rho^0 - \mathcal{L}^*\big) + \frac{2\,\mathbb{E}\big[\|\mathbf{e}^1\|^2\big]}{\gamma_K \cdot K} + \frac{\displaystyle\sum_{k=0}^{K-1} 4\gamma_k^2\sigma^2}{\gamma_K \cdot K} + \frac{1}{K}\sum_{k=0}^{K-1}\delta_k + \left[\frac{1}{\eta_K \cdot K} + \frac{16\eta_1 L_f^2}{\gamma_K \cdot K}\right]\sum_{k=0}^{K-1}\hat{\delta}_k \tag{4.15}$$

Lemma 8 (Violation Bound) *Under the same conditions as Lemma 3, with $\delta_k = c/(k+1)$ for some $c > 0$, we have*

$$\frac{1}{T}\sum_{k=0}^{T-1}\mathbb{E}\big[\|\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}^{k+1})\|^2 + \|[\boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}^{k+1})]_+\|^2\big] = \mathcal{O}\left(\frac{\log T}{T}\right) \tag{4.16}$$

Proof 1 *Let $r_k := \boldsymbol{\lambda}^k - \boldsymbol{\mu}^k$. By the $\boldsymbol{\lambda}$-update (3.9), the equality block satisfies $r_{\mathcal{E}}^{k+1} = \rho\,\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}^{k+1})$, while the inequality block obeys*

$$r_{\mathcal{I}}^{k+1} = \big[\boldsymbol{\mu}_{\mathcal{I}}^{k+1} + \rho\,\boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}^{k+1})\big]_+ - \boldsymbol{\mu}_{\mathcal{I}}^{k+1}.$$

Since $\boldsymbol{\mu}_{\mathcal{I}}^{k+1} \geq \mathbf{0}$ by the initialization and the convex combination update (3.6), we have $[c_i(\boldsymbol{x}^{k+1})]_+ \leq |r_i^{k+1}|/\rho$ for every $i \in \mathcal{I}$, and hence

$$\|\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}^{k+1})\|^2 + \|[\boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}^{k+1})]_+\|^2 \leq \frac{1}{\rho^2}\|r_{k+1}\|^2, \tag{4.17}$$

so it suffices to bound $\frac{1}{T}\sum_{k=0}^{T-1}\mathbb{E}[\|r_{k+1}\|^2]$.

Step 1: Derive the recursion for $\|r_{k+1}\|^2$**.** *By the triangle inequality and* $(a+b)^2 \le 2(a^2+b^2)$*,*

$$\|r_{k+1}\|^2 \le 2\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2 + 2\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^{k+1}\|^2. \tag{4.18}$$

Applying Lemma 2 (4.5) to the second term and (4.6) to the first:

$$\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^{k+1}\|^2 = (1-\theta_k)^2\|r_k\|^2,$$

$$\mathbb{E}[\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2 \mid \mathcal{F}_k] \le 2\rho^2 L_c^2\, \mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \frac{\rho^2\delta_k^2}{2}.$$

Substituting into (4.18) and using $(1-\theta_k)^2 \le (1-\theta_k)$ *(since* $\theta_k \in (0,1]$*):*

$$\mathbb{E}[\|r_{k+1}\|^2 \mid \mathcal{F}_k] \le 2(1-\theta_k)\|r_k\|^2 + 4\rho^2 L_c^2\, \mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \rho^2\delta_k^2. \tag{4.19}$$

Step 2: Sum the recursion. *Rearranging (4.19):*

$$2\theta_k\|r_k\|^2 \le 2\|r_k\|^2 - \mathbb{E}[\|r_{k+1}\|^2 \mid \mathcal{F}_k] + 4\rho^2 L_c^2\, \mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \rho^2\delta_k^2. \tag{4.20}$$

Taking total expectation and summing over $k = 0, \dots, T-1$*:*

$$2\sum_{k=0}^{T-1} \theta_k\, \mathbb{E}[\|r_k\|^2] \le 2\sum_{k=0}^{T-1} \mathbb{E}[\|r_k\|^2] - \sum_{k=0}^{T-1} \mathbb{E}[\|r_{k+1}\|^2] + 4\rho^2 L_c^2 \sum_{k=0}^{T-1} \mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \rho^2 \sum_{k=0}^{T-1} \delta_k^2. \tag{4.21}$$

Note that the first two sums on the right telescope:

$$2\sum_{k=0}^{T-1} \mathbb{E}[\|r_k\|^2] - \sum_{k=0}^{T-1} \mathbb{E}[\|r_{k+1}\|^2] = \sum_{k=0}^{T-1} \mathbb{E}[\|r_k\|^2] - \sum_{k=1}^{T-1} \mathbb{E}[\|r_k\|^2] + \mathbb{E}[\|r_0\|^2] - \mathbb{E}[\|r_T\|^2] \le \mathbb{E}[\|r_0\|^2]. \tag{4.22}$$

Substituting (4.22) into (4.21):

$$2\sum_{k=0}^{T-1} \theta_k\, \mathbb{E}[\|r_k\|^2] \le \|r_0\|^2 + 4\rho^2 L_c^2 \sum_{k=0}^{T-1} \mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \rho^2 \sum_{k=0}^{T-1} \delta_k^2. \tag{4.23}$$

Dividing both sides of (4.23) by T*:*

$$\frac{2}{T}\sum_{k=0}^{T-1} \theta_k\, \mathbb{E}[\|r_k\|^2] \le \frac{\|r_0\|^2}{T} + \frac{4\rho^2 L_c^2}{T} \sum_{k=0}^{T-1} \mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \frac{\rho^2}{T} \sum_{k=0}^{T-1} \delta_k^2. \tag{4.24}$$

5 Numerical Experiments

A Supported Lemmas

Lemma 9 (Vector Azuma-Hoeffding Inequality) *Let* $\{\boldsymbol{v}^k\}_{k=1}^K$ *be a vector-valued martingale difference sequence adapted to filtration* $\{\mathcal{F}_k\}_{k=0}^K$*, satisfying*

$$\mathbb{E}[\boldsymbol{v}^k \mid \mathcal{F}_{k-1}] = \mathbf{0}, \qquad \|\boldsymbol{v}^k\| \le a_k, \quad k = 1, \dots, K. \tag{A.1}$$

Then for any $\delta \in (0,1)$*, with probability at least* $1-\delta$*,*

$$\left\|\sum_{k=1}^K \boldsymbol{v}^k\right\| \le 3\sqrt{\log(1/\delta)\sum_{k=1}^K a_k^2}. \tag{A.2}$$

The above bound is a standard vector-valued Azuma-Hoeffding concentration result.

References

[1] Ahmet Alacaoglu, Olivier Fercoq, and Stephen J. Wright. Convergence of first-order algorithms for meta-learning with moreau envelopes. *arXiv preprint*, 2023. Verify: this may be Alacaoglu–Wright (2022) on stochastic primal-dual with STORM for fully-stochastic constrained optimization.

[2] Ahmet Alacaoglu and Stephen J. Wright. Complexity of single loop algorithms for nonlinear programming. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, Proceedings of Machine Learning Research, 2024. arXiv:2311.00678.

[3] Yossi Arjevani, Yair Carmon, John C. Duchi, Dylan J. Foster, Nathan Srebro, and Blake Woodworth. Lower bounds for non-convex stochastic optimization. *Mathematical Programming*, 199:165–214, 2023.

[4] Digvijay Boob, Qi Deng, and Guanghui Lan. Stochastic first-order methods for convex and nonconvex functional constrained optimization. *Mathematical Programming*, 197(1):215–279, 2023.

[5] Digvijay Boob, Qi Deng, and Guanghui Lan. Feasible and objective convergence of a first-order method for nonconvex and nonsmooth constrained optimization. *SIAM Journal on Optimization*, 34:2180–2211, 2024. Verify publication year and volume.

[6] Coralia Cartis, Nicholas I. M. Gould, and Philippe L. Toint. On the complexity of steepest descent, Newton's and regularized Newton's methods for nonconvex unconstrained optimization problems. *SIAM Journal on Optimization*, 20(6):2833–2852, 2010. Use Cartis–Gould–Toint 2014 exact penalty paper if available.

[7] Ying Cui, Bo Tang, and Yanli Liu. Complexity of an inexact proximal-point penalty method for constrained smooth non-convex optimization. *Computational Optimization and Applications*, 82:175–224, 2022. Verify exact authors and title.

[8] Ying Cui, Bo Tang, and Yanli Liu. An adaptive exact penalty method for nonconvex stochastic equality-constrained optimization. *Mathematical Programming*, 2024. Verify exact authors, title, and publication details.

[9] Frank E. Curtis, Daniel P. Robinson, and Baoyu Shi. A stochastic sequential quadratic programming algorithm for nonlinear-equality-constrained optimization with rank-deficient Jacobians. *Mathematics of Operations Research*, 46(4):1236–1267, 2021.

[10] Ashok Cutkosky and Francesco Orabona. Momentum-based variance reduction in non-convex SGD. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[11] Cong Fang, Chris Junchi Li, Zhouchen Lin, and Tong Zhang. SPIDER: Near-optimal non-convex optimization via stochastic path-integrated differential estimator. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[12] Yifan Huang et al. Stochastic augmented Lagrangian with fixed penalty for nonconvex optimization with linear inequality constraints. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. Verify exact authors and title; uses Moreau envelope, $\mathcal{O}(\varepsilon^{-3})$ for linear inequality constraints.

[13] Zhuoxu Jia and Benjamin Grimmer. First-order constraint-qualification-free optimality conditions and algorithms for nonconvex constrained optimization. In *Proceedings of the 40th International Conference on Machine Learning*, 2023. Verify exact venue and title.

[14] Zhiyan Jin and Tianhao Wang. A single-loop smoothed gradient descent-ascent algorithm for nonconvex-concave min-max problems with coupled linear constraints. In *Advances in Neural Information Processing Systems*, 2022. Verify exact title, authors, and venue; may be different paper.

[15] Zichong Li, Pin-Yu Chen, Sijia Liu, Songtao Lu, and Yangyang Xu. Rate-improved inexact augmented Lagrangian method for constrained nonconvex optimization. 130:2170–2178, 2021. Published as @inproceedings; verify details.

[16] Zichong Li, Pin-Yu Chen, Sijia Liu, Songtao Lu, and Yangyang Xu. Stochastic inexact augmented Lagrangian method for nonconvex expectation constrained optimization. *Computational Optimization and Applications*, 87:117–147, 2024.

[17] Sijia Liu and Yangyang Xu. A first-order algorithmic framework for distributionally robust logistic regression. *arXiv preprint*, 2021. Verify: this corresponds to Liu–Xu exact penalty + SPIDER paper for nonsmooth fully-stochastic constrained optimization, achieving $\mathcal{O}(\varepsilon^{-4})$ and $\mathcal{O}(\varepsilon^{-6})$ complexities.

[18] Haihao Lu and Song Mei. Single-loop algorithms for stochastic nonconvex functional constrained optimization. *arXiv preprint*, 2022. Verify exact authors, title, and publication venue.

[19] Zhi-Quan Luo and Jong-Shi Pang. Error bounds for analytic systems and their applications. *Mathematical Programming*, 67(1):1–28, 1994.

[20] Shiqian Ma, Lingxiao Hou, and Mladen Kolar. Proximally constrained methods for weakly convex optimization with weakly convex constraints. *INFORMS Journal on Optimization*, 4(3):246–270, 2022. Verify exact title and venue.

[21] Sen Na, Mihai Anitescu, and Mladen Kolar. Inequality constrained stochastic nonlinear optimization via active-set sequential quadratic programming. *Mathematical Programming*, 196:743–791, 2022.

[22] Chuwen Shi et al. A momentum-based linearized augmented Lagrangian method for nonconvex constrained stochastic optimization. *Mathematical Methods of Operations Research*, 2025. Verify exact authors and title; achieves $\mathcal{O}(\varepsilon^{-3.5})$ with dual momentum.

[23] Chuwen Shi and Runcheng Lin. A linearized augmented Lagrangian method with variance reduction for nonconvex stochastic constrained optimization. *arXiv preprint arXiv:2211.XXXXX*, 2022. Verify exact authors, title, and publication venue.

[24] Yue Xie and Stephen J. Wright. Complexity of proximal augmented Lagrangian for nonconvex optimization with nonlinear equality constraints. *Journal of Scientific Computing*, 86(3):38, 2021.

[25] Yifan Zhang and Zhi-Quan Luo. A proximal alternating direction method of multiplier for linearly constrained nonconvex minimization. *SIAM Journal on Optimization*, 33(3):2046–2078, 2023. Verify exact title, volume, and year; the key result is $\mathcal{O}(\varepsilon^{-2})$ for linear constraints with constant penalty via Luo–Pang error bound.